MOHAMED ALAA

Unity Engineer | Unity Developer

CONTACT

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Egypt, Cairo, 1st settlement

PORTFOLIO

https://mohamedalaa2180.github.io/Alaa-portfolio/

EDUCATION

Bachelor of software engineering
University of Ain Shams | 2014 - 2018

PROFILE

Senior Unity Engineer with 7+ years of hands-on expertise designing, building, and launching 2D, 3D, AR, and VR solutions using Unity. Led engineering teams at Genesis Creations through July 2026, delivering cutting-edge VR training platforms and multiplayer titles.

Proven track record with 600K+ combined downloads across published titles & key projects:

- Kortifo, multiplayer 2D card battle game with 200,000+ downloads
- Experience Makkah, cross-platform VR simulation with 250,000+ downloads
- Rehlatie, educational 2D mobile game with 150,000+ downloads
- VRC & DXB, VR sterile training platform & 3D operations facility tour
- Gaming Room VR, playable Meta Quest VR room with custom physics

Skilled in multiplayer systems, VR/AR mechanics, performance optimization, and scalable cloud architectures using Unity, C#, Photon/Netcode, Meta SDK, and Firebase.

ACTIVITIES

- Joined as game development mentor in traverse summit event 2024 the biggest high schoolers summit in mena region
- Unity developer instructor in Coursera
- Unity Asset Store Publisher – creating and publishing reusable tools and game development assets for the Unity developer community.

WORK EXPERIENCE

Lead Unity Engineer

May 2025 - July 2026

Genesis Creations

- Started by leading a team of 4 mid-level and senior Unity developers, then expanded to leading juniors, mid-seniors, and seniors across multiple concurrent projects.
- Collaborated with product owners to define sprint goals, break down features into tasks, and plan agile sprints effectively.
- Provided technical mentorship to junior developers across multiple teams, supporting their growth and maintaining code standards.
- Conducted training sessions for bootcamp Unity developers, teaching intermediate and advanced Unity concepts to accelerate their learning and integration.
- Improved company-wide development workflow by introducing modern technologies, tools, and strategic process optimizations.
- Designed scalable development architectures for new projects and refactored legacy codebases to improve maintainability and performance.

Senior Game Developer

APRIL 2024 - May 2025

Nahdet Misr

- Contributed to the development of **Rehlatie** an educational **2D mobile game using Unity**, working closely with other developers and testers to ensure feature completeness and code quality.
- Designed and implemented a **performance optimization strategy** that reduced app size and improved runtime efficiency by using **Unity Addressables** and refactoring complex systems.
- Integrated **Firebase** services including **Cloud Functions**, **Remote Config**, **Crashlytics**, and **Analytics** to enable real-time configuration and improve stability.
- Managed **builds and deployment processes** for both **App Store** and **Google Play**, ensuring compliance with store guidelines and maintaining release stability.
- Provided **technical support** and conducted **code reviews** for junior team members to uphold code quality and promote best development practices.
- **Rehlatie achieved over 150,000 downloads across both stores**, highlighting the success of its design, performance, and user engagement.

Senior Game Developer

OCT 2022 - APRIL 2024

UMAMI Games

- Worked on **Kortifo**, a multiplayer 2D card game, contributing to both core gameplay mechanics and backend integration using **Unity**.
- Designed and architected the **Daily and Weekly Objectives System**, integrating with Unity Cloud Code to enable dynamic and remotely configurable mission logic.
- Built a fully featured **Shop System** supporting virtual currency and real-money **In-App Purchases (IAP)**, leveraging **Unity IAP** and **Remote Config** for scalable offer management.
- Developed a **Card Effect Queue System** to determine execution order for complex in-game interactions, ensuring consistent and fair gameplay logic.
- Implemented **Unity Netcode for GameObjects** to support real-time multiplayer gameplay, handling synchronization and networked card actions.
- Integrated **Unity services** including **Cloud Save**, **Remote Config**, **Cloud Code**, and **IAP**, enabling robust backend connectivity and remote feature tuning.
- Collaborated directly with the art team to translate **Figma** designs into polished in-game UI with high visual fidelity and user-centric UX.
- Supported the deployment and maintenance of builds on **App Store** and **Google Play**, contributing to submission workflows and QA validation.
- **Kortifo achieved over 200,000 downloads across both stores**, reflecting the game's polished execution and strong market engagement.

. NET DEVELOPER

MAR 2022 - JULY 2022

EDGE PRO

- Contributed to the backend development of several **government web platforms** using **ASP.NET** and **C#**, with a focus on API development and system integration.
- Designed and implemented **RESTful APIs** to support secure data exchange between services and external systems.
- Worked with **MySQL** and **PostgreSQL** databases to handle data modeling, query optimization, and stored procedures.
- Collaborated with front-end developers and system administrators to ensure smooth deployment and functionality across all modules.
- Assisted in debugging and maintaining legacy code, improving stability and system responsiveness.

Junior Game Developer

JAN 2022 - MAR 2022

AVATARIS

- Worked remotely on a mobile game project, primarily focusing on **UI development** and integration using **Unity**.
- Translated designs from **Figma** into interactive and responsive in-game interfaces, ensuring alignment with gameplay flow.
- Collaborated with graphic designers to optimize asset import workflows and improve rendering quality across devices.
- Maintained clean and scalable UI architecture to support future feature expansions and localization.

Junior Game Developer

OCT 2019 - DEC 2021

Vhorus

- Served as the **sole Unity developer** in a 3-person team alongside one artist and one graphic designer.
- Developed multiple **Augmented Reality (AR)** applications primarily for **advertising and interactive brand activations**, using Vuforia and AR Foundation with Unity.
- Designed and built a **2D Tangram Puzzle Game**, focusing on intuitive user interaction and mobile performance.
- Led the full development of **Experience Makkah**, a **cross-platform VR simulation** of the Hajj journey. using **Google Cardboard SDK**. The app launched on both **Android** and **iOS**, and surpassed **250,000 downloads** on the Google Play Store.
- Implemented scene management, interactive camera systems, and spatial audio to deliver a deeply immersive mobile VR experience.
- Optimized build size, memory usage, and rendering pipeline for smooth operation on mid-tier mobile devices.

Junior Game Programmer

FEB 2019 - OCT 2019

CLOUD SOFT

- Acted as the **sole Unity developer**, responsible for end-to-end development of **2D educational mobile games** incorporating **Augmented Reality (AR)** features.
- Designed and implemented interactive learning experiences tailored for students, blending AR content with traditional gameplay mechanics.
- Utilized **Unity** and AR toolkits to develop engaging educational content optimized for Android devices.
- Managed asset integration, gameplay logic, UI/UX flow, and performance tuning to ensure accessibility and stability across a range of mobile devices.
- Delivered projects independently, aligning with educational objectives and meeting production timelines with minimal supervision.

CORE SKILLS

- Game Engines: Unity (2D, 3D), Addressables System, Universal Render Pipeline (URP)
- Multiplayer: Unity Netcode for GameObjects, Photon PUN, Photon Fusion
- AR / VR: Meta Quest / Oculus SDK, XR Interaction Toolkit, OpenXR, Vuforia, AR Foundation
- Backend / Live Ops: Firebase (Crashlytics, Remote Config, Cloud Code), Unity Cloud Save, IAP
- Languages: C#, ASP.NET
- Tools & DB: Figma, Git, Postman, MySQL, PostgreSQL, Jira, Trello, UniTask
- Platforms: Android, iOS, Meta Quest (VR), Windows PC

KEY PROJECTS

VRC - Virtual Interactive Cleanroom | Meta Quest (VR) | Team Leader & Lead Developer | Genesis Creations

- Led 7 developers building Canada's first VR sterile compounding training platform for hospital pharmacies and technicians.
- Engineered dynamic cleaning system for any tool/surface, automated workflow orchestration, and real-time protocol compliance.

DXB - Interactive 3D Facility Walkthrough | Meta Quest (VR) & Windows PC | Lead Unity Engineer | Genesis Creations

- Architected multi-zone interactive 3D facility tour for GITEX demos using Meta XR SDK, OpenXR, and Cinemachine.
- Integrated Schneider Electric NetEco APIs via typed UniTask layer, driving live telemetry dashboards (power, cooling, alarms).

Gaming Room VR | Meta Quest (VR) | Solo Developer | Showcase Project

- Engineered two-handed pool cue mechanics with cue-axis strike assist physics and angle-tested stickable darts release system.
- Calibrated stylized hand meshes directly driven by Meta Interaction SDK hand skeleton across custom bone layouts.

Kortifo - Multiplayer Football Card Game | Android & iOS (200,000+ Downloads) | Senior Game Developer | UMAMI Games

- Implemented real-time multiplayer networking with Unity Netcode for GameObjects and networked card action synchronization.
- Architected Daily & Weekly Objectives with Unity Cloud Code, Card Effect Queue System, and real-money/virtual IAP shop.

Experience Makkah - Virtual Hajj Journey | Android & iOS (250,000+ Downloads) | Sole Unity Developer | Vhorus

- Handled entire development lifecycle: Google Cardboard SDK integration, spatial audio, and interactive camera systems.
- Optimized rendering pipeline, draw calls, and memory footprint for smooth 60 FPS performance on mid-tier mobile devices.

Rehlatie - Educational Mobile Game | Android & iOS (150,000+ Downloads) | Senior Game Developer | Nahdet Misr

- Devised and executed Addressables asset streaming architecture, significantly decreasing initial store download size.
- Integrated Firebase suite (Cloud Functions, Remote Config, Crashlytics, Analytics) for remote feature tuning and live analytics.

Darbk Khdr - Saudi Desert Adventure Game | Meta Quest (VR) & Mobile | Sole Developer | Saudi Cultural Heritage

- Built complete gameplay loop from concept to deployment featuring character selection and desert adventure exploration.
- Optimized VR camera systems, asset pipelines, and locomotion mechanics for comfortable immersion on Meta Quest.